

Observation Theory: Geometry, Distortion, and Reliability as Properties of Observation

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Abstract

We develop Observation Theory, the thesis that operational geometry, distortion, and reliability are properties not of an object alone but of the relation among the object, an observer, a channel, and a resolution budget. An observer is a consumer, a map from a representation to a task output, with a metric on that output. A second-order expansion of the downstream distortion isolates one controlling quantity: the reconstruction error read through the consumer, the trace of the error second moment weighted by a positive semidefinite read operator, the pullback of the output metric along the consumer. Ordinary reconstruction is the isotropic corner where this operator is the identity, the home of the Gaussian mean squared error specialization of Shannon rate distortion and of Gauss least squares. We prove that the read operator is the pullback geometry of distinguishability, inducing a quotient of the representation by what the consumer cannot separate; that it is recoverable from black box consumer queries under a specified observation distribution; and that it defines a consumer relative rate distortion function whose achievability and converse we establish for a general source, with a coding theorem for the true nonlinear consumer divergence that reduces exactly to output coding. We instantiate the read operator for attention, gradient optimization, and retrieval, and position it as the identifiable geometric middle term between Shannon rate distortion and the information bottleneck. Every empirical claim is governed by a sealed prior registration with a preset bar; four predictions are confirmed and one is refuted.

Index Terms

Rate–distortion theory, source coding, quantization, information geometry, distinguishability, indirect source coding, information bottleneck, neural network compression, reproducibility.

I. INTRODUCTION

Shannon fixed a distortion measure and asked for the fewest bits meeting it [1], [2]; Gauss fixed the same measure and asked for the least-squares estimate [3]. This paper reads their common assumption as the variable of interest: the error that matters is not the one a code makes but the one a consumer *reads*, and different consumers read different subspaces of the same error. Once the distortion is that of the task, Euclidean reconstruction is one corner—the isotropic observer $P_C = I$ —and any anisotropic or rank-deficient P_C reads the error differently.

The idea that one should code a source to serve a *functional* of it, not the source, is not new: it is the indirect (remote) rate–distortion problem [5], [6], [7]; its lossless, discrete form is coding for computing and graph entropy [8], [9]; its modern face is the rate–distortion–perception tradeoff [10] and task-oriented / semantic communication [11]; and its deployed form, for neural networks, is loss-aware (Hessian-weighted) compression, from Optimal Brain Damage/Surgeon [16], [17] to Hessian-aware quantization and GPTQ [18], [19]. What is missing across these is a single, *identifiable geometric* object that names the task’s metric and lets one predict, recover, and test it. Observation Theory supplies it: the read operator P_C , the consumer’s local output metric, sitting as the geometric middle term between Shannon’s Gaussian Euclidean-MSE specialization ($P_C = I$) and the information bottleneck [12] (an arbitrary, non-geometric relevance variable). We contribute (i) the map placing these literatures on one axis; (ii) P_C ’s *identifiability*—recovery from black-box consumer queries under a specified observation distribution, validated blind against a planted ground truth, which the quantization literature (estimating the Hessian from calibration data) never validates against truth; (iii) an ex-ante extreme-value reliability certificate; (iv) budget-relativity of the verdict; and (v) a registration-first empirical program—sealed predictions, ex-ante bars, a refuted

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Companion software, the sealed pre-registrations, and the claim ledger that govern every empirical result are publicly available at <https://github.com/ahb-sjsu/geometric-observation>. A continuous-integration workflow re-runs the reproducible experiments and verifies the theoretical constructions on each commit.

TABLE I
OBSERVATION THEORY AS THE IDENTIFIABLE GEOMETRIC MIDDLE TERM BETWEEN TWO CLASSICAL EXTREMES. THE READ OPERATOR P_C IS THE AXIS.

Framework	P_C	Character
Shannon rate–distortion (Gaussian MSE)	I (isotropic)	fixed, task-agnostic; the reconstruction corner where Gauss least squares lives
Observation Theory	$J^\top G J$	geometric, positive semidefinite, water-fillable, and <i>identifiable</i> from the consumer
Information bottleneck	—	an arbitrary, non-geometric relevance variable

prediction reported as prominently as the confirmations—whose recurring demonstration is a single code whose downstream verdict *inverts* when nothing changes but the consumer. The three companion works are Paper I (the angular observer and the spectral budget-inversion mechanism) [20], Paper II (geometric key quantization and the reconstruction-is-not-the-target negative) [21], and the claim ledger [22]; this is their information-theoretic synthesis.

II. THE CONTROLLING QUANTITY

Definition 1 (Instance). *An instance is $(X, \mathcal{C}, \mathcal{Q})$: a representation with vectors $x \in \mathbb{R}^d$, a consumer $\mathcal{C} : \mathbb{R}^d \rightarrow \mathbb{R}^m$ with a divergence D_Y on its outputs, and a family $\mathcal{Q}$ of lossy codes producing $\hat{x} = x + \delta$. The downstream distortion is the pointwise random quantity $d_{\mathcal{O}} := D_Y(\mathcal{C}(x), \mathcal{C}(\hat{x}))$, measured on the consumer output, never assumed from the ambient metric of X ; write $\bar{d}_{\mathcal{O}} := \mathbb{E}[d_{\mathcal{O}}]$ for its expectation over the instance.*

Example 1 (A running instance). A concrete instance to keep in mind is a single attention head. Here X is the set of key vectors, the consumer $\mathcal{C}$ is the softmax-attention map $x \mapsto \text{softmax}(q^\top x / \sqrt{d})$ that the head’s queries q induce, the output divergence D_Y is the Kullback–Leibler divergence between attention distributions, and $\mathcal{Q}$ is a family of low-bit key quantizers. Then $d_{\mathcal{O}}$ measures how far a quantizer moves the head’s attention, and—as §III derives—the read operator turns out to be the query second moment $P_C = \mathbb{E}[qq^\top]$: a code is judged only by how its error projects onto the directions the queries actually read. We return to this instance when instantiating each result.

Proposition 1 (Consumer-projected distortion). *Let D_Y be twice differentiable with $D_Y(u, u) = 0$, $\nabla_v D_Y(u, v)|_{v=u} = 0$, and local output metric $G(u) = \nabla_v^2 D_Y(u, v)|_{v=u} \succeq 0$ (the Fisher information metric when D_Y is the KL divergence or a kindred statistical divergence, a generic local metric otherwise); let $J(x) = \partial \mathcal{C} / \partial x$. For error $\delta = \hat{x} - x$ with conditional second moment $\Sigma_\delta(x) = \mathbb{E}[\delta \delta^\top \mid x]$, the leading-order expected distortion is*

$$\bar{d}_{\mathcal{O}} = \frac{1}{2} \mathbb{E}_x \text{tr}(P_C(x) \Sigma_\delta(x)) + r, \quad P_C(x) = J(x)^\top G(\mathcal{C}(x)) J(x) \succeq 0,$$

so P_C is the pullback of the local output metric G along $\mathcal{C}$. Making the asymptotics explicit: for a small-noise family $\delta_\varepsilon = \varepsilon Z$ with $\mathbb{E}\|Z\|^2 < \infty$ and the second-order Taylor remainder dominated uniformly on a neighborhood of the diagonal, $\bar{d}_{\mathcal{O}} = \frac{\varepsilon^2}{2} \mathbb{E}[Z^\top P_C(X) Z] + o(\varepsilon^2)$ (i.e. $r = o(\varepsilon^2)$). When $\Sigma_\delta(x)$ is uncorrelated with the point-dependent weight $P_C(x)$ this factors as $\frac{1}{2} \text{tr}(P_C \Sigma_\delta)$ with $P_C = \mathbb{E}_x[P_C(x)]$, $\Sigma_\delta = \mathbb{E}_x[\Sigma_\delta(x)]$. The controlling object is the second moment, not the covariance: pointwise $\text{tr}(P_C(x) \Sigma_\delta(x)) = \text{tr}(P_C(x) \text{Cov}[\delta \mid x]) + \mu_\delta(x)^\top P_C(x) \mu_\delta(x)$ with $\mu_\delta(x) = \mathbb{E}[\delta \mid x]$, then averaged over x , so a biased code’s offset is part of the distortion, not averaged away—which is why quantizer bias is first-class (§III). Reconstruction distortion $\mathbb{E}\|\delta\|^2 = \text{tr}(\Sigma_\delta)$ is the isotropic corner $P_C = I$.

Proof. Expand $\mathcal{C}(x + \delta) = \mathcal{C}(x) + J\delta + O(\|\delta\|^2)$ and D_Y to second order about its minimum; the value and gradient terms vanish, leaving $\frac{1}{2} \delta^\top (J^\top G J) \delta + o(\|\delta\|^2)$. Take $\mathbb{E}_x$; the factored form requires the stated uncorrelatedness (for real quantizers both J and Σ_δ depend on x , so the exact object is the x -averaged trace). $\square$

We keep $\mathbb{E}_x \text{tr}(P_C(x) \Sigma_\delta(x))$ as primary and use the factored $\text{tr}(P_C \Sigma_\delta)$ as the convenience it is. (We drop the harmless $\frac{1}{2}$.)

Remark (When the factored form is exact). Because the rest of the paper leans on $\text{tr}(P_C \Sigma_\delta)$, it is worth stating when it holds. The factorization is *exact* whenever the error second moment $\Sigma_\delta(x)$ does not co-vary with the read

operator $P_C(x)$ across the support—in particular for a *linear* consumer, where P_C is constant. It holds to good approximation for a *fixed-codebook* quantizer whose per-coordinate error statistics are roughly homogeneous over x (uniform or per-channel key quantization on a representation whose read subspace is stable), which covers the codes studied in §IV. When the consumer is strongly point-dependent—a deep network whose $P_C(x)$ swings with the activation—the coupled expectation $\mathbb{E}_x \text{tr}(P_C(x)\Sigma_\delta(x))$ must be retained; Appendix A then gives the exact, unfactored treatment.

Distinguishability, and the observer's quotient

The form $\delta^\top P_C(x)\delta = (J\delta)^\top G(J\delta)$ is not, at root, a compression cost; it is a *distinguishability*. It measures how far the consumer's outputs $\mathcal{C}(x)$ and $\mathcal{C}(x+\delta)$ separate in the output metric G , and when G is the Fisher information metric of the output distribution, $\delta^\top P_C\delta$ is the local statistical distinguishability (the leading KL / χ^2) of those outputs. So P_C is the pullback through $\mathcal{C}$ of the geometry of distinguishability: Observation Theory is pullback geometry.

Two consequences follow, and both recur below. First, define the global operational equivalence $x \sim_{\mathcal{C}} x' \iff D_Y(\mathcal{C}(x), \mathcal{C}(x')) = 0$ (statistical indistinguishability of the outputs). Its infinitesimal shadow is $\ker P_C(x) = \ker(G^{1/2}J)$, which equals $\ker J$ when $G \succ 0$ on $\text{im } J$ but is larger when G is only semidefinite—the output can move parametrically yet stay indistinguishable. Under a constant-rank / integrability condition on $P_C(x)$ this null distribution integrates to leaves tangent to $\ker P_C(x)$; that the *global* divergence vanishes along each leaf—so the leaves are exactly the fibers of $\sim_{\mathcal{C}}$ —needs a further compatibility condition, namely $D_Y(\mathcal{C}(x), \mathcal{C}(x')) = 0$ along them (equivalently, $\mathcal{C}$ factors through $X/\sim_{\mathcal{C}}$). Where it holds, the consumer's true object is the quotient $X/\sim_{\mathcal{C}}$, not X : perturbations in $\ker P_C$ exist in the representation but have *no operational consequence*. In the fixed- P_C model of §VIII this quotient is represented by ΠX and Prop. 7 gives the zero-distortion rate $R_C(0) = H(\Pi X)$; the point-dependent case is settled unconditionally in Appendix A (Cor. 3: $R_C(0) = H(\mathcal{C}(X))$), via the true divergence, needing none of the compatibility conditions above—rate is expended only on distinctions the consumer can resolve. Second, $\ker P_C$ is gauge-like *but only relative to $\mathcal{C}$* : another consumer may read those directions, so it is not a true gauge redundancy (which would be invisible to every admissible observer). One observer's noise is another's signal; one observer's state variable is another's gauge freedom. Adjoining the budget of §VII, an observer is a triple $\mathcal{O} = (\mathcal{C}, G, B)$ —readout, output metric, and resolution—and the geometry it resolves is relative to all three.

Principle of observational relativity. Operational fidelity, distortion, and resolved geometry are not properties of the object alone; they are properties of the relation among object, observer $(\mathcal{C}, G)$, channel, and budget, and are ill-posed until all are specified. Observation induces geometry; each observer pulls back a distinct operative geometry and quotient.

The rest of the paper derives P_C for concrete consumers, shows it recoverable from queries, and studies what the channel (§VI) and budget (§VII) do to the geometry it induces.

III. DERIVATION OF THE READ OPERATOR

Proposition 2 (Softmax attention). *Read keys $\{k_s\}_{s=1}^S$ through a query bank: logits $l_s = q^\top k_s / \sqrt{d}$, output $p = \text{softmax}(l)$, $D_Y = \text{KL}$. Under key perturbation δ_s ,*

$$\bar{d}_{\mathcal{O}} = \frac{1}{2d} \mathbb{E}_q \text{Var}_s^p(q^\top \delta_s) + O(\|\delta\|^3) = \frac{1}{2d} \mathbb{E}_q \text{tr}(qq^\top A_q) + O(\|\delta\|^3), \quad A_q = \sum_s p_s(q) \tilde{\delta}_s \tilde{\delta}_s^\top,$$

where $\tilde{\delta}_s = \delta_s - \sum_{s'} p_{s'} \delta_{s'}$ is the across-token p -demeaned error and A_q its $p(q)$ -weighted second moment. This factors to $\frac{1}{2d} \text{tr}(Q \bar{\Sigma}_\delta)$, with $Q = \mathbb{E}_q[qq^\top]$ and $\bar{\Sigma}_\delta = \mathbb{E}_q[A_q]$, only when A_q is uncorrelated with $qq^\top$ (the proviso of Prop. 1); in general the expectation stays coupled, since p depends on q .

Proof. The Fisher metric of softmax is $G = \text{diag}(p) - pp^\top$, so $\text{KL}(p \parallel \text{softmax}(l + \Delta l)) = \frac{1}{2} \text{Var}_s^p(\Delta l_s) + O(\Delta l^3)$ with $\Delta l_s = q^\top \delta_s / \sqrt{d}$. The $-pp^\top$ term is the softmax's invariance to a common logit shift; it removes the p -mean, giving $\tilde{\delta}_s$. Since p depends on q , the outer $\bar{\Sigma}_\delta$ carries the $\mathbb{E}_q$ and the final Q -factorization inherits the uncorrelatedness proviso of Prop. 1. $\square$

TABLE II

THE READ OPERATOR P_C INSTANTIATED FOR THREE CONSUMERS. ORDINARY RECONSTRUCTION IS THE ISOTROPIC CORNER $P_C = I$.

Consumer	Output divergence D_Y	Read operator P_C	Recovered from
Softmax attention	Kullback–Leibler	query second moment $\mathbb{E}[qq^\top]$ (coupled expectation)	finite-difference Jacobian probe of the softmax output
Gradient optimizer	quadratic step error	Hessian H (Gauss–Newton curvature if nonconvex)	calibration curvature estimate
Retrieval ranking	margin / rank loss	subspace projector $\mathbb{E}[qq^\top]$	query-bank second moment
Reconstruction	squared Euclidean	identity I (isotropic corner)	— (task-agnostic)

Two consequences are the phenomena we later measure. The feature-space factor is generated by $qq^\top$, reducing to $Q = \mathbb{E}_q[qq^\top]$ under the decorrelation / uniform-weight regime above; so an isotropic bank yields an isotropic feature operator (reconstruction-like) while a subspace bank projects—the read operator, hence the winning code, changes with the query distribution. And only the *demeaned* error survives, so the across-token mean is a nuisance the softmax cannot see. This last fact resolves an apparent tension with the DC-offset collapse of Paper II [21]: a truly common error component is free by shift invariance, but a per-channel DC *offset* is not a common component—it inflates the quantizer’s range, wasting levels and enlarging the *visible* per-token error, which is exactly why an asymmetric zero-point (asym-NF4) repairs it at no bit cost. Invisible-to-the-consumer and free-to-encode are the same statement; range waste is a different one.

Proposition 3 (Gradient-descent optimizer; exact). *Let $L(\theta) = \frac{1}{2}\theta^\top H\theta$ with $H \succeq 0$, $g = H\theta$, and $\hat{g} = g + \delta$ with conditionally unbiased error $\mathbb{E}[\delta \mid \theta] = 0$, used in a step of size η . Then $L(\theta - \eta\hat{g}) - L(\theta - \eta g) = \frac{1}{2}\eta^2 \delta^\top H\delta - \eta \delta^\top H(\theta - \eta g)$, whose error-quadratic part is $\frac{1}{2}\eta^2 \text{tr}(H\delta\delta^\top)$; the linear cross term vanishes in expectation ($\mathbb{E}[\delta \mid \theta] = 0$), giving $\mathbb{E}[d\mathcal{O}] = \frac{1}{2}\eta^2 \text{tr}(H\Sigma_\delta)$, so $P_C = H \succeq 0$ exactly.*

This is loss-aware pruning/quantization made a metric: Hessian-weighted deletion [16], [17], [18], [19] is $\text{tr}(H\Sigma_\delta)$, and calibration-based Hessian estimation is the production form of the probe of §V. Reconstruction is the flat case $H = I$. For a non-convex loss the Hessian may be indefinite; the metric reading then takes P_C to be the PSD Gauss–Newton / Fisher curvature in place of H .

Remark (Retrieval). For score-ranking $s(q, x_i) = q^\top x_i$, the inverted-pair fraction is, to leading order for smooth rank statistics, monotone in $\text{tr}(P_C\bar{\Sigma}_\delta) / \text{tr}(P_C\bar{\Sigma}_0)$, with $P_C = \mathbb{E}[qq^\top]$, $\bar{\Sigma}_\delta = \text{Cov}_i(\delta_i)$ the across-item error covariance and $\bar{\Sigma}_0 = \text{Cov}_i(x_i)$ the signal covariance (both demeaned across items); P_C is a subspace projector when queries concentrate. Rank statistics are non-smooth; we treat this instance empirically (§IV).

IV. DISTORTION: THE CONSUMER DETERMINES THE METRIC (GO-2)

Proposition 2 predicts that at matched reconstruction the ranking of two codes is set by $\text{tr}(P_C\Sigma_\delta)$, hence by the query distribution.

a) Dissociation.: Two 4-bit codes at an audited matched budget (spread 0.19 bit) with *identical* reconstruction (0.0938 vs 0.0934): the whole-vector code is $2.53\times$ worse on softmax-KL (12/12 seeds); a wrong-quotient control is $21\times$ worse at $\sim 5\times$ the reconstruction. Reconstruction is not the controlling quantity ([demonstrated]).

b) Flip.: Holding reconstruction fixed and changing only the query bank, the ranking *inverts*: isotropic $\Rightarrow$ code *a* better ($2.5\times$, 12/12); subspace $\Rightarrow$ code *b* better (12/12). The directly computed $\text{tr}(P_C\Sigma_\delta)$ tracks the sign on every seed in both consumers; reconstruction, a consumer-blind scalar, cannot (rank correlation 0.40 vs 1.00). This is Proposition 2 with Q swapped ([demonstrated]).

c) Replication and out-of-sample.: The flip replicates on retrieval ranking (different representation, set-valued consumer): identical reconstruction (0.0964), ranking flips with the read subspace (log-advantage -4.70 vs $+4.65$, 12/12), $\text{tr}(P_C\Sigma_\delta)$ tracking, reconstruction blind ([replicated]). It generalizes to optimization with $P_C = H$ (Prop. 3): at matched reconstruction the curvature-metric distortion flips between two Hessians (12/12), tracked

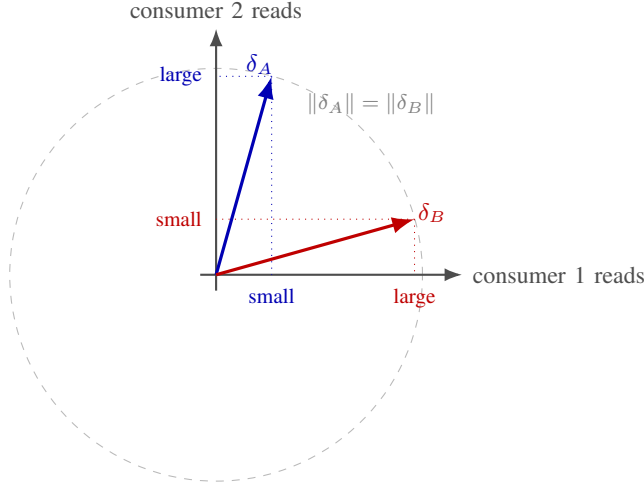


Fig. 1. The flip. Two lossy codes produce errors δ_A, δ_B of *equal* reconstruction norm (dashed circle), so reconstruction cannot separate them. Consumer 1 (horizontal read operator) sees a small δ_A projection and a large δ_B one, and prefers code A ; consumer 2 (vertical read) prefers code B . The downstream verdict $\text{tr}(P_C \Sigma_\delta)$ reverses with the read operator while reconstruction remains blind—the characteristic signature of Observation Theory.

by $\text{tr}(H \Sigma_\delta)$, reconstruction uninformative (-0.20)—and under $H = I$ reconstruction tracks *perfectly* (1.00), the derived $P_C = I$ corner observed ([predicted], out-of-sample). One derivation; three read operators. These consumers are *synthetic*—the point is to validate the theory’s logic (the $H = I$ positive control being the clean confirmation); the trained-model hardening (§XI), attempted prospectively, initially missed on a non-reconstruction-matched design and is then *cleared* by a corrected, reconstruction-matched rematch on a real Llama attention layer.

V. IDENTIFIABILITY: THE READ SUBSPACE IS RECOVERABLE (GO-1)

Proposition 4 (Recovery of the read subspace). *Let $\mathcal{C}(x) = f(Wx)$, $W \in \mathbb{R}^{r \times d}$ of rank r , f coordinatewise with $\mathbb{E}[f'_i(Wx)^2] > 0$ for every coordinate i (no saturated unit). Then $\mathbb{E}_x[J^\top J] = W^\top \mathbb{E}[\text{diag}(f'^2)]W$ has rank r with column space $\text{row}(W)$, so its top- r eigenvectors span the read subspace; a finite-difference estimate $\hat{J}u \approx (\mathcal{C}(x+hu) - \mathcal{C}(x-hu))/2h$ recovers it from consumer queries alone.*

We built such a probe—receiving only the callable $\mathcal{C}(\cdot)$ and the dimensions, never W , the codes, or d_O (blinding enforced in code). It recovers the planted read subspace at overlap 0.936 against a 0.059 chance baseline ($\approx 16\times$), and the *blind* $\text{tr}(\hat{P}_C \Sigma_\delta)$ predicts the flip of §IV on 12/12 seeds in both consumers with rank correlation 1.00 —identical to ground truth—while reconstruction cannot (0.40) ([predicted]). Two scope notes. Prop. 4 recovers the read *subspace* for the $f(Wx)$ model with a Euclidean output metric; recovering the *full* operator $\mathbb{E}_x[J^\top G J]$ —its eigenvalues, and a general G —is the broader program. And P_C depends on the probe-point distribution and on G , so it is identifiable from black-box consumer queries *under a specified observation distribution*, not from the callable alone. Loss-aware compression estimates such a metric from calibration data; the contribution here is validating *recovery against a planted ground truth under enforced blinding*, and composing it: from consumer queries under that distribution one predicts which code wins, before compressing anything.

VI. CAPACITY: A DERIVED VACUITY THRESHOLD (GO-3)

Proposition 5 (Retrieval vacuity threshold). *Model the $N-1$ distractor scores, standardized to mean 0, variance 1, as i.i.d. standard Gaussian (a reference model). Since $\Pr[M_{N-1} \leq m] = \Phi(m)^{N-1}$, top-1 success $\Pr[\hat{\mu} > M_{N-1}]$ crosses $\frac{1}{2}$ at the exact median of the maximum,*

$$\mu_{\text{crit}} := \text{med}(M_{N-1}) = \Phi^{-1}(2^{-1/(N-1)}), \quad \text{leading asymptotic } \sqrt{2 \ln(N-1)},$$

where $\hat{\mu} = (\overline{\text{true}} - \overline{\text{distr}})/\text{std}$ is the margin certificate and $\rho = \hat{\mu}/\mu_{\text{crit}}$; the certificate is vacuous for $\rho < 1$.

Derivation. $\Pr[M_{N-1} \leq m] = \Phi(m)^{N-1}$; setting it to $\frac{1}{2}$ gives $m = \Phi^{-1}(2^{-1/(N-1)})$, the exact median. The Gumbel centering is $\alpha_N = \sqrt{2 \ln(N-1)} - b_N(\ln \ln(N-1) + \ln 4\pi)/2$ with scale $b_N = 1/\sqrt{2 \ln(N-1)}$; the crude

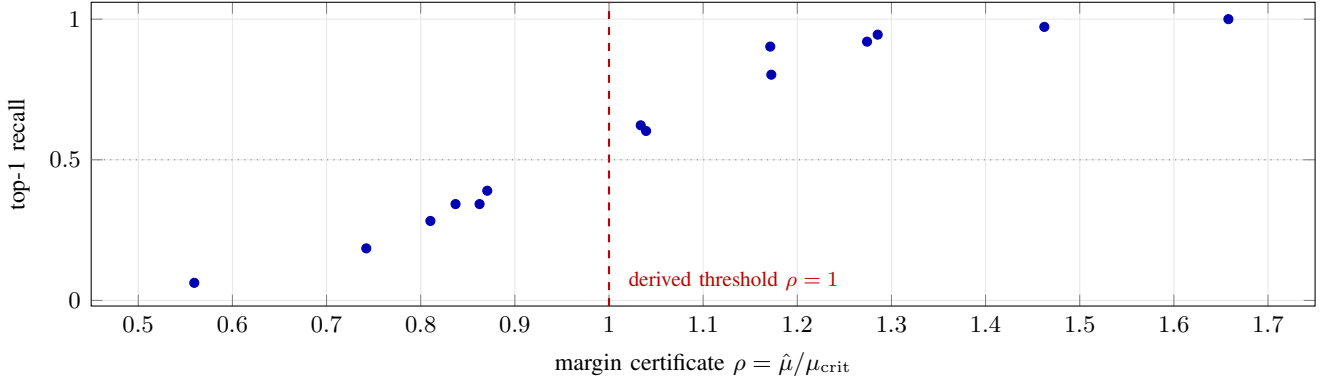


Fig. 2. Capacity (GO-3), from the committed result artifact. The derived margin certificate ρ orders top-1 retrieval recall across 14 structurally distinct corpora at Spearman correlation 0.99; the recall- $\frac{1}{2}$ crossing sits at $\rho \approx 0.95$, near the extreme-value vacuity threshold $\rho = 1$ (dashed) at which the channel is predicted to die *before* any retrieval is run.

$\sqrt{2 \ln(N-1)}$ omits the $-b_N(\ln \ln(N-1) + \ln 4\pi)/2$ correction and so *overshoots*. Median and mean share this centering and differ at order b_N : $\text{med}(M_{N-1}) = \alpha_N - b_N \ln \ln 2 + o(b_N)$ and $\mathbb{E}[M_{N-1}] = \alpha_N + \gamma b_N + o(b_N)$ (γ Euler–Mascheroni), so the mean exceeds the median by $(\gamma + \ln \ln 2) b_N$. The recall- $\frac{1}{2}$ crossing is the exact median $\Phi^{-1}(2^{-1/(N-1)})$ —not the mean, not the leading order. $\square$

Across 14 structurally distinct corpora the certificate orders them by recall at rank correlation 0.991, beating the un-normalized margin (0.873), with a finite-width transition ([demonstrated]). The registered harness used the *mean* $\mathbb{E}[M_{N-1}]$ (Monte-Carlo) as its μ_{crit} reference, and the recall- $\frac{1}{2}$ crossing sits at $\rho = 0.948$ —slightly *below* 1 against that reference. This is expected and quantitative: the exact $\frac{1}{2}$ threshold is the *median*, below the mean by $\Theta(1/a_N)$, so the $\sim 5\%$ shortfall is largely the mean–median offset (against the exact median of this proposition the crossing sits at ≈ 1). Positive dependence among distractors *can* lower it further by reducing the effective count; heavier tails, by contrast, *raise* the maximum—the two must not be grouped. The exact $\frac{1}{2}$ crossing further assumes a *fixed* standardized true margin and i.i.d. Gaussian distractors; when the true score is itself random or correlated with the distractors, ρ is a reference certificate rather than an exact recall threshold. The *form* of the threshold is nonetheless robust: the $\sqrt{2 \ln N}$ scaling is the centering of the Gumbel max-domain of attraction, shared by every distractor law with sub-exponential (lighter-than-polynomial) tails—Gaussian, exponential, Weibull-type—so non-Gaussianity *within* that domain moves the constants (α_N, b_N) , not the scaling law; positive correlation enters as an *effective count*, replacing N by θN through the extremal index $\theta \in (0, 1]$ and shifting μ_{crit} by $\frac{\ln \theta}{2\sqrt{2 \ln N}}$ at leading order; and only genuinely heavy (Fréchet-domain, polynomial) tails change the scaling itself—the certificate’s stated failure mode. The 14-corpus ordering above ($\rho = 0.99$ on clustered, non-Gaussian real embeddings) is the empirical counterpart: real structure shifts the absolute death point but preserves the ordering the certificate is used for. Capacity says, from a computable quantity, *where* the channel dies before any retrieval is run.

Using the certificate. For a practitioner, ρ is a cheap pre-flight check: estimate the standardized true-versus-distractor margin on a small calibration set, divide by the derived μ_{crit} , and read off ρ . A value below 1 warns that single-stage retrieval is at or past its vacuity point—budget for a re-ranking stage or a larger index before deployment—while ρ comfortably above 1 certifies headroom. Because ρ preserves the recall *ordering* across corpora, it also ranks candidate corpora or index configurations *without* running full retrieval.

VII. RATE: THE OPERATIVE VERDICT DEPENDS ON THE BUDGET (GO-4)

The read operator, capacity, and distortion are stated at a fixed observer budget; the budget is a rate, and it inverts the verdict. The mechanism is spectral: an eigenfunction φ_k with eigenvalue λ_k oscillates on the wavelength $\lambda_k^{-1/2}$, and Weyl’s law $N(\lambda) \sim c \text{Vol}(M) \lambda^{d_M/2}$ (d_M the intrinsic dimension) grows the count of modes below a fixed scale with resolution.

Lemma 1 (Fixed-band consistency). *At a fixed low-mode budget m , as the sample grows the graph operator converges spectrally to the manifold operator [14], so the fixed low- λ band sharpens and geodesic fidelity converges toward its continuum value and is predicted to improve asymptotically (the spectral-convergence mechanism of*

Paper I [20]). Convergence does not force step-wise monotonicity in N ; the monotone finite-sample trend below is the registered empirical result, not a consequence of the lemma.

Holding instead the resolved scale τ fixed (spectral-gap matching), $m = N(\tau)$ grows with N ; the newly resolved modes near τ have wavelength approaching the sampling scale and, for a non-ideal substrate, inject sub-geodesic detail, so geodesic fidelity is nonincreasing—Paper I’s registered inversion [20]. We replicate it on a new substrate: 768-dimensional multilingual book-paragraph embeddings (Project Gutenberg translations; 32,000 points, three seeds). Fixed $m = 10$: geometry preservation rises with N (trend $+0.4, +1.0, +1.0$). Spectral-gap-matched, $m : 10 \rightarrow 121/126/159$: geometry *collapses* (trend -1.0 every seed); the budgets agree at the smallest N (gap ≤ 0.002) ([replicated]). Opposite verdicts about the same data, set by the observer’s rate: “clean geometry” is a relation between observer resolution and substrate scale, not a substrate invariant.

VIII. A CONSUMER-RELATIVE RATE-DISTORTION FUNCTION

The consumer-projected distortion is the single-letter measure $d_C(x, \hat{x}) = (x - \hat{x})^\top P_C(x - \hat{x})$ for a fixed P_C (constant, or the average $\mathbb{E}_x[P_C(x)]$ under the proviso of Prop. 1); define $R_C(D) = \inf\{I(X; \hat{X}) : \mathbb{E} d_C(X, \hat{X}) \leq D\}$. This fixed- P_C model must be kept distinct from the *local* observation metric $P_C(x)$ of Prop. 1: for a nonlinear consumer $P_C(x)$ varies, there is no single tilted source—a direction invisible at one point can be visible at another—and the coding theorem for that case is open. The results below, and Appendix A, are for the fixed weighted-quadratic model, whose theory is classical [4].

Proposition 6 (Gaussian rate–distortion; Berger reduction). *For $X \sim \mathcal{N}(0, \Sigma_x)$, in the basis diagonalizing $P_C^{1/2} \Sigma_x P_C^{1/2}$ with eigenvalues γ_i ,*

$$R_C(D) = \sum_i \frac{1}{2} \log_2^+ \frac{\gamma_i}{\theta}, \quad \sum_i \min(\gamma_i, \theta) = D,$$

a reverse water-filling on the P_C -weighted source spectrum. For $P_C = I$ this is Shannon’s Gaussian $R(D)$ [2]; the estimation dual is Gauss–Markov least squares [3].

Proof. $\text{tr}(P_C \Sigma_\delta) = \|P_C^{1/2}(x - \hat{x})\|^2$; with $y = P_C^{1/2}x$ this is MSE on $y \sim \mathcal{N}(0, P_C^{1/2} \Sigma_x P_C^{1/2})$, and Shannon water-filling applies. The transform is invertible off $\ker P_C$, on which the consumer is blind and no rate is spent. $\square$

Bits water-fill on directions the consumer reads *and* the source varies, spending nothing on $\ker P_C$ —why a reconstruction-optimal code ($P_C = I$ allocation) is beaten by a consumer-matched one when $P_C \neq I$, and why the winner *flips* with P_C (§IV). The $H = I$ control confirms it: only at the isotropic $P_C = I$ is reconstruction the right objective, and there it is exactly right.

Proposition 7 (Consumer-lossless floor; discrete source). *Let X be discrete with finite entropy, Π the projector onto $\text{range}(P_C)$. There exist codes with $\text{tr}(P_C \Sigma_\delta) = 0$ at rate*

$$R_C(0) = H(\Pi X) \leq H(X), \quad H(X) - R_C(0) = H(X | \Pi X) \geq 0,$$

with equality iff Π is injective on $\text{supp}(X)$; the saving is strict exactly when distinct symbols differ only within the consumer-invisible subspace $\ker P_C$.

Proof. Achievability: $\text{tr}(P_C \Sigma_\delta)$ depends on δ only through $\Pi \delta$; sending ΠX losslessly and reconstructing $(I - \Pi)X$ by its conditional mode gives $\Pi \delta = 0$, zero distortion, at rate $H(\Pi X)$. *Converse:* zero distortion forces $\Pi \hat{X} = \Pi X$ a.s., so $I(X; \hat{X}) \geq I(\Pi X; \Pi \hat{X}) = H(\Pi X)$. Hence $R_C(0) = H(\Pi X)$; if Π collapses two support points, $H(\Pi X) < H(X)$. $\square$

Remark (Continuous sources). When the tilted source $Y = P_C^{1/2}X$ has a nondegenerate continuous component, lossless-to-the-consumer coding costs infinite rate ($R_C(D) \rightarrow \infty$ as $D \rightarrow 0$, consistent with Prop. 6); if X lies entirely in $\ker P_C$ then Y is degenerate and the floor is zero. The surviving statement is that at *every* $D > 0$ the water-filling of Prop. 6, equivalently the reduction of Cor. 1, spends zero rate on $\ker P_C$: the operational rate depends only on the r -dimensional tilted source. This lossless-to-the-consumer phenomenon is exactly coding for computing and graph entropy in the discrete world [8], [9], here given a geometric read operator.

These form a *conceptual axis*, not a formal nesting: Shannon rate–distortion ($P_C = I$, the isotropic observer), consumer-relative rate–distortion (P_C a fixed geometric, *identifiable* operator, Props. 6,7), and the information

bottleneck [12]—which optimizes a different object (a stochastic relevance variable) and to which the consumer-relative term connects only through the local-Gaussian relevance limit ($Y = \mathcal{C}(X)$ deterministic, second-order). We claim no containment theorem. The middle term is the operational one— $P_{\mathcal{C}}$ is PSD, water-fillable, and recoverable from the consumer (Prop. 4)—and its coding theorem (achievability and converse, fixed $P_{\mathcal{C}}$, general source) is Appendix A.

IX. A BOUNDARY: THE REFUTED DENSITY QUOTIENT (GO-5)

Not every plausible face survives a sealed prediction. In the spectral setting the $\alpha=1$ anisotropic normalization of a diffusion operator cancels sampling density and recovers density-free geometry [13]: there the density mode lies in $\ker P_{\mathcal{C}}$. We conjectured this density quotient likewise restores the invariant in a *non-spectral* domain (locally-scaled ADC retrieval, a hubness correction [15]). It does not, across four prospective tests (direct affinity twice, ADC on real embeddings, non-spectral diffusion-distance). The reason is Proposition 6: for a direct inner-product consumer the density is *read*—it is not in $\ker P_{\mathcal{C}}$ —so quotienting it removes signal and manufactures anti-hubs; in the diffusion domain the direction is right ($\alpha=1$ optimal, recovered non-spectrally) but the effect is under 2% and *not density-specific* (a uniform-density control gains as much). Per its sealed registration the conjecture is [refuted]. The sharpened statement—the $\alpha=1$ cancellation is a property of the diffusion operator and does not transfer to direct observation—is Proposition 6 read backwards: rate is saved on $\ker P_{\mathcal{C}}$ only when the nuisance actually lies there. The refutation sharpens the theory rather than undermining it.

X. NEGATIVE RESULTS AND METHODOLOGY

Twelve standing negatives bound the theory and are reported with the prominence of the positives. Reconstruction cosine is not a proxy for key quality (Paper II [21], the motivating negative). The flip is observable only for reconstruction-*matched* codes: absent a tie one code Pareto-dominates and reconstruction is not falsified (a prospective miss locating a precondition, consistent with the water-filling of Prop. 6). The projected-variance trace governs the discriminating comparison but is not a complete rank statistic over all codes. The density quotient is treated in §IX. Two negatives are methodological self-corrections: in two instances a substantive claim passed at 12/12 while an over-strict secondary gate we had specified failed, and in two others a degenerate substrate was detected only by a regime-coverage gate.

Every empirical claim was governed by a dated, hashed registry entry committed—sealed—before the first measurement it governs; bars were fixed ex ante; misses enter the ledger as misses; amendments only dated and logged before unblinding; prospective claims (§V, VII, IX) were blinded where feasible. The house rule is that the paper may not assert what the ledger cannot show.

XI. DISCUSSION

In summary, Observation Theory identifies operational geometry with the pullback geometry of distinguishability and makes that geometry recoverable and experimentally consequential. It implies a change of default: where a conventional pipeline minimizes reconstruction and presumes that the task follows, the theory instead prescribes recovering the consumer’s read operator (§V), water-filling bits on $P_{\mathcal{C}}$ rather than on I (§VIII), respecting the channel’s vacuity threshold (§VI), and recognizing that the verdict is relative to the observation budget (§VII). Against the literatures it unifies—indirect source coding, coding for computing, rate-distortion-perception, task-oriented communication, loss-aware compression—the contribution is making the task’s metric a single identifiable geometric object and testing it adversarially. The decisive validation is not conceptual but empirical, and we conducted it on a real model. A first preregistered blind probe on a Llama-3.2-3B attention layer (Gate B, GO-P-2026-020) *missed* its sealed bar: recovery was above chance but sub-threshold (0.567), and its registered quantizer pair was *not* reconstruction-matched, so reconstruction predicted the winner for free (16/16) and the projected predictor could not *beat* it—the same precondition that governs the synthetic dissociation. The corrected rematch (GO-P-2026-021) fixes exactly this: two key-error arms tied in reconstruction to 7.5×10^{-9} but differing downstream (median relative divergence gap 1.85), with an improved probe. On the trained model it *clears all four sealed gates*—the blind $\text{tr}(\hat{P}_{\mathcal{C}}\Sigma_{\delta})$ predicts the reconstruction-invisible worse arm on 16/16 heads while reconstruction, provably tied, agrees on only 2/16 (its own floating-point noise), and the read operator is recovered at overlap 0.647. The

TABLE III

THE FALSIFIABLE CORE. EACH ROW IS A SEALED, BEFORE-THE-FACT REGISTRATION WITH AN EX-ANTE ACCEPTANCE BAR; FOUR ARE BORNE OUT AND ONE IS REFUTED AND CARRIED AS A BOUNDARY. A PROSPECTIVE TRAINED-MODEL (LLAMA-3.2-3B) HARDENING OF GO-2 FIRST MISSED ON A NON-RECONSTRUCTION-MATCHED DESIGN AND IS THEN CLEARED BY A CORRECTED REMATCH (§XI).

Claim	Prediction	Class	Result
GO-1 identifiability	blind probe recovers the read subspace	predicted	overlap 0.94 vs 0.06 chance; predicts the flip 12/12
GO-2 distortion	a reconstruction-identical code flips its verdict with the consumer	demonstrated, replicated	flips on attention, retrieval, and optimization
GO-3 capacity	extreme-value vacuity threshold locates retrieval death	demonstrated	orders 14 corpora at $\rho = 0.99$
GO-4 rate	a fixed-budget verdict inverts under budget matching	replicated	inverts on real 768-d embeddings
GO-5 density quotient	$\alpha=1$ restores the invariant non-spectrally	refuted	4 prospective misses; boundary

theory’s mechanism thus operates on a trained frontier layer, called *before* compression by a blind probe; the GO-P-2026-020 miss stands as an honest negative on the code design it exposed. Open theory: we settle the fixed- P_C achievability and converse of $R_C(D)$ in Appendix A and lift it to a fully point-dependent *surrogate* metric (Cor. 2), and Appendix A settles the *true* nonlinear divergence too—its single-letter theorem at every D (Thm. 2) and its exact reduction to output-side coding, the tilt being $\mathcal{C}$ itself (Thm. 3), with the quotient floor unconditional (Cor. 3). What remains open is narrower: the curved-image high-resolution case, the multi-observer rate region (distinct quotients, no single reduction), the consumer-relative *channel* coding theorem behind §VI, and whether the flip is exactly R_C ’s argmin migrating with P_C .

The coding-theoretic core of these claims lives in the appendices, and a reader focused on the information theory should read them next. Appendix A proves the achievability and converse of the consumer-relative rate–distortion function $R_C(D)$ for a general source; Appendix A then extends the theorem to the *true* nonlinear consumer divergence and shows that the problem reduces *exactly* to ordinary source coding of the consumer output (the read-operator “tilt” being $\mathcal{C}$ itself). These are the results that make “distortion is what the consumer reads” a coding theorem rather than a heuristic, and they are load-bearing for every rate claim in the body.

Underlying all of this is a single principle: an object has no unique operational geometry; each observer $\mathcal{O} = (\mathcal{C}, G, B)$ pulls back one, and the quotient $X / \sim_{\mathcal{C}}$ is the domain on which it acts. Four of the five registered predictions are confirmed, one is refuted, and twelve standing negatives are retained; the framework is now positioned within the established literature.

APPENDIX

Throughout the main theorem, P_C is a *fixed* PSD operator (constant, or $\mathbb{E}_x[P_C(x)]$ under the proviso of Prop. 1); Cor. 2 then lifts the operational statement to a point-dependent $P_C(x)$ surrogate metric.

Lemma 2 (Reduction to the tilted source). *Let $Y = P_C^{1/2}X$, on $\text{range}(P_C)$ of dimension $r = \text{rank } P_C$. Then $d_C(x, \hat{x}) = \|y - \hat{y}\|^2$ with $\hat{y} = P_C^{1/2}\hat{x}$, and $R_C(D) = R_Y(D)$, the MSE rate–distortion function of Y ; $\ker P_C$ costs zero rate.*

Proof. $d_C = \|P_C^{1/2}(x - \hat{x})\|^2 = \|y - \hat{y}\|^2$. Objective and constraint depend on $\hat{X}$ only through $\hat{Y}$, so by data processing $I(X; \hat{X}) \geq I(Y; \hat{Y})$; the minimizer places all information in Y (a channel $p(\hat{y} | y)$, then $\hat{X} = P_C^{-1/2}\hat{Y}$ on the range, conditional mean on $\ker P_C$), attaining equality. Hence the minimum is $R_Y(D)$. $\square$

Theorem 1 (Achievability and converse). *Let X take values in a standard Borel space, $\{X_t\}$ i.i.d. $\sim p_X$, with a measurable reconstruction alphabet and a reference letter $\hat{x}_0$ satisfying $\mathbb{E} d_C(X, \hat{x}_0) < \infty$ and $\mathbb{E} d_C(X, \hat{x}_0)^2 < \infty$.*

For every $D > 0$: (i) for any $R > R_C(D)$ there are $(2^{nR}, n)$ codes with $\limsup_n \mathbb{E} \frac{1}{n} \sum_t d_C(X_t, \hat{X}_t) \leq D$; (ii) every code of expected distortion $\leq D$ has rate $\geq R_C(D)$; and R_C is convex and nonincreasing.

Proof. This is the single-letter rate–distortion theorem for a memoryless source on a standard Borel space with a measurable difference distortion and finite reference-letter moment [4]; we recall it in *information-density* form, which uses no differential entropies and so holds on general alphabets. *Achievability.* Fix $\varepsilon > 0$ and a test channel $p^*(\hat{x} | x)$ with $\mathbb{E} d_C \leq D$ and $I(X; \hat{X}) \leq R_C(D) + \varepsilon$; let $p_{\hat{X}}$ be the output marginal and $\imath(x; \hat{x}) = \log \frac{dP_{X\hat{X}}}{d(P_X \times P_{\hat{X}})}$ the information density. Draw $\{\hat{X}^n(w)\}_{w=1}^{2^{nR}}$ i.i.d. $\sim \prod_t p_{\hat{X}}$; encode x^n by any w with $\frac{1}{n} \sum_t \imath(x_t; \hat{X}_t(w)) \leq I(X; \hat{X}) + \varepsilon$ and $\frac{1}{n} \sum_t d_C(x_t, \hat{X}_t(w)) \leq D + \varepsilon$, else declare failure and reconstruct with the constant $\hat{x}_0^n$. Let $\mathcal{T}$ be the joint event $\{\frac{1}{n} \sum_t \imath \leq I + \varepsilon, \frac{1}{n} \sum_t d_C \leq D + \varepsilon\}$; by the weak law $P_{X\hat{X}}^n(\mathcal{T}) \rightarrow 1$, so for all but an $o(1)$ -probability set $\mathcal{G}_n$ of source sequences the conditional probability $P_{\hat{X}^n|X^n=x^n}(\mathcal{T}) \geq 1 - o(1)$. Changing measure from $P_{\hat{X}^n|X^n}$ to the codebook marginal $P_{\hat{X}^n} = \prod_t p_{\hat{X}}$ on $\mathcal{T}$ costs at most $2^{n(I+\varepsilon)}$ (there the density $\prod_t \frac{dP_{\hat{X}|X}}{dP_{\hat{X}}} = 2^{\sum_t \imath} \leq 2^{n(I+\varepsilon)}$), so for each $x^n \in \mathcal{G}_n$ a single random codeword passes the encoding test with conditional probability $\geq 2^{-n(I+2\varepsilon)}(1 - o(1))$. Hence $\Pr[\text{fail} | x^n] \leq (1 - 2^{-n(I+2\varepsilon)}(1 - o(1)))^{2^{nR}} \rightarrow 0$ for $R > I + 2\varepsilon$, uniformly over $x^n \in \mathcal{G}_n$, and $\Pr[\text{fail}] \leq \Pr[\mathcal{G}_n^c] + \sup_{x^n \in \mathcal{G}_n} \Pr[\text{fail} | x^n] \rightarrow 0$. On success the distortion is $\leq D + \varepsilon$; on failure it is $\frac{1}{n} \sum_t d_C(X_t, \hat{x}_0)$, whose contribution to the expectation is $\leq \sqrt{\mathbb{E}[(\frac{1}{n} \sum_t d_C(X_t, \hat{x}_0))^2]} \sqrt{\Pr[\text{fail}]} \rightarrow 0$ by Cauchy–Schwarz, finite by the second-moment hypothesis on $\hat{x}_0$. Thus $\limsup_n \mathbb{E} \frac{1}{n} \sum_t d_C \leq D + \varepsilon$; let $\varepsilon \rightarrow 0$. *Converse.* With $D_t = \mathbb{E} d_C(X_t, \hat{X}_t)$, $nR \geq H(W) \geq I(X^n; \hat{X}^n) \geq \sum_t I(X_t; \hat{X}_t) \geq \sum_t R_C(D_t) \geq nR_C(\frac{1}{n} \sum_t D_t) \geq nR_C(D)$, by memorylessness, the definition of R_C , Jensen (convexity), and monotonicity. Convexity and monotonicity follow from time-sharing and distortion relaxation. $\square$

Corollary 1. $R_C(D)$ equals the MSE rate–distortion of the r -dimensional tilted source $Y = P_C^{1/2}X$; for Gaussian X it is Prop. 6; discretely, $R_C(0) = H(\Pi X)$ (Prop. 7). The operational rate to serve a consumer is governed entirely by the source projected into the read subspace—the precise sense in which Shannon’s theorem is the corner $P_C = I$, where the read subspace is everything.

Corollary 2 (Point-dependent surrogate). The coding theorem holds verbatim for the point-dependent quadratic distortion $d_{P_C}(x, \hat{x}) = (\hat{x} - x)^\top P_C(x)(\hat{x} - x)$ whenever $P_C(\cdot)$ is measurable with $\|P_C(x)\| \leq \Lambda < \infty$ and $\mathbb{E}\|X\|^4 < \infty$. The achievability and converse of Thm. 1 nowhere use that the distortion is a difference distortion—only that it is a nonnegative measurable single-letter distortion with a finite second-moment reference letter, and $d_{P_C}(x, \hat{x}_0) \leq \Lambda\|\hat{x}_0 - x\|^2$ supplies both. Hence the operational rate–distortion function $R_C(D) = \inf\{I(X; \hat{X}) : \mathbb{E} d_{P_C}(X, \hat{X}) \leq D\}$ exists and is achievable for a fully point-dependent read operator.

What the fixed- P_C specialization keeps and the point-dependent case loses is therefore sharp: (i) the closed-form reverse water-filling (Cor. 1) needs a single linear tilt $Y = P_C^{1/2}X$, whereas with $P_C(x)$ the tilt $x \mapsto P_C(x)^{1/2}x$ is nonlinear and R_C has no closed form; (ii) the exact floor $R_C(0) = H(X/\sim_C)$ requires the integrability/compatibility conditions of §II to identify the quotient; and (iii) d_{P_C} is the leading-order surrogate for the true divergence $D_Y(\mathcal{C}(x), \mathcal{C}(\hat{x}))$, so coding to it certifies the true downstream distortion only in the high-rate (small- D) regime. The genuinely open problem is thus not the existence of $R_C(D)$ under a point-dependent metric—Cor. 2 settles that—and, as Appendix A shows, not the true nonlinear divergence either: Thm. 2 gives its single-letter theorem at every D , and Thm. 3 its exact reduction (the tilt is $\mathcal{C}$ itself, with no linear surrogate needed). What remains open is narrower, and stated in §B.4.

Here X is i.i.d. $\sim p_X$ on a Polish space $\mathcal{X}$, reconstruction alphabet $\hat{\mathcal{X}} = \mathcal{X}$, $\mathcal{C} : \mathcal{X} \rightarrow \mathcal{U}$ Borel into a Polish output space, and $D_Y : \mathcal{U} \times \mathcal{U} \rightarrow [0, \infty]$ jointly Borel with $D_Y(u, u) = 0$. The true distortion is $d_C(x, \hat{x}) := D_Y(\mathcal{C}(x), \mathcal{C}(\hat{x}))$ and $R_C^{\text{true}}(D) := \inf\{I(X; \hat{X}) : \mathbb{E} d_C(X, \hat{X}) \leq D\}$. The three results settle, in turn, existence of the single-letter theorem at non-vanishing D , the exact nonlinear tilt, and the quantitative relation to the quadratic surrogate d_{P_C} of Cor. 2. The load-bearing identities (Thm. 3, Cor. 3) and the high-resolution formula (Thm. 4) were confirmed to machine precision by an independent Blahut–Arimoto / reverse-water-filling check committed with the ledger [22].

A. Existence: the theorem ports verbatim

Theorem 2 (Single-letter theorem for the true divergence). If there is $\hat{x}_0$ with $\mathbb{E} d_C(X, \hat{x}_0)^2 < \infty$, then for every $D > 0$, $R_C^{\text{true}}(D)$ is the operational rate–distortion function (achievability for any $R > R_C^{\text{true}}(D)$; converse), and

R_C^{true} is convex and nonincreasing.

Proof. d_C is a nonnegative, jointly Borel, single-letter distortion (a composition of Borel maps). The information-density achievability and converse of Thm. 1 use only nonnegativity, joint measurability, memorylessness, and the reference-letter second moment; they nowhere use symmetry, the triangle inequality, difference structure, or $d(x, \hat{x}) = 0 \iff x = \hat{x}$ —as Cor. 2 records. The proof applies verbatim with d_C for d_{P_C} [23]. Nothing about the nonlinearity of $D_Y \circ (\mathcal{C}, \mathcal{C})$ obstructs the classical theorem; the existence half of the “open problem” was mislocated. $\square$

B. The exact nonlinear tilt: reduction to the output source

Theorem 3 (Output reduction; the tilt is $\mathcal{C}$). *Let $U := \mathcal{C}(X)$, $V := \mathcal{C}(\hat{X}) \subseteq \mathcal{U}$, and $R_U(D) := \inf\{I(U; \hat{U}) : \hat{U} \in V \text{ a.s., } \mathbb{E} D_Y(U, \hat{U}) \leq D\}$ the ordinary rate–distortion function of the output source U under D_Y , reconstruction alphabet V . With D_Y jointly Borel and a universally measurable right inverse $s : V \rightarrow \hat{\mathcal{X}}$ of $\mathcal{C}$ (automatic for Borel $\mathcal{C}$ between Polish spaces: V is analytic and Jankov–von Neumann uniformization supplies s [24]), $R_C^{\text{true}}(D) = R_U(D)$ for every $D \geq 0$.*

Proof. ($\geq$) Any admissible $\hat{X}$ gives $\hat{U} := \mathcal{C}(\hat{X}) \in V$ with $\mathbb{E} D_Y(U, \hat{U}) = \mathbb{E} d_C \leq D$, and $I(X; \hat{X}) \geq I(U; \hat{X}) \geq I(U; \hat{U})$ by data processing (U a function of X ; $\hat{U}$ a function of $\hat{X}$). ($\leq$) Given a test channel $p(\hat{u} | u)$ with $I(U; \hat{U}) \leq R_U(D) + \varepsilon$, draw $\hat{U} | X \sim p(\cdot | \mathcal{C}(X))$ and set $\hat{X} := s(\hat{U})$; then $\mathcal{C}(\hat{X}) = \hat{U}$, so $\mathbb{E} d_C = \mathbb{E} D_Y(U, \hat{U}) \leq D$, and $X \rightarrow U \rightarrow \hat{U} \rightarrow \hat{X}$ is Markov, giving $I(X; \hat{X}) \leq I(X; \hat{U}) = I(U; \hat{U})$. The equality uses *both* $I(U; \hat{U} | X) = 0$ (U is a deterministic function of X) and $I(X; \hat{U} | U) = 0$ (the Markov property), so $I(X; \hat{U}) = I(X, U; \hat{U}) = I(U; \hat{U})$. The universally measurable s defines the joint law of $(X, \hat{X})$; on the standard Borel product this disintegrates into a Borel conditional kernel, so the exhibited channel is feasible (universal measurability is used only to build the joint law). Because $d_C(x, \hat{x})$ depends on $\hat{x}$ only through $\mathcal{C}(\hat{x}) \in V$, restricting reconstructions to $\{s(\hat{u})\}$ loses nothing—the reduction is tight. Let $\varepsilon \rightarrow 0$. $\square$

Corollary 3 (Quotient floor). *If D_Y is a strict divergence ($D_Y(u, u') = 0 \iff u = u'$) and X is discrete with finite entropy, then $R_C^{\text{true}}(0) = H(\mathcal{C}(X)) = H(X/\sim_C)$. These two hypotheses are the only ones: none of the integrability, constant-rank, or leaf-compatibility conditions the surrogate needs (§II, Cor. 2) arises, because the true divergence defines the quotient globally rather than infinitesimally. (If D_Y is degenerate, replace $\mathcal{C}(X)$ by its quotient under D_Y -indistinguishability.) Prop. 7 is the linear case $\mathcal{C} = \Pi$.*

Corollary 4 (Order of operations). *The reduction of Thm. 3 removes all dependence on $\ker P_C$, quotienting it exactly prior to any linearization; the surrogate’s pathologies (singular weights, non-integrable null distributions) are artifacts of linearizing before quotienting. The “nonlinear tilt” has a closed-form reduction—it is $\mathcal{C}$ itself; what has no closed form is a tilt constrained linear in x , which the true problem never needed. Residual regularity is then governed not by a $\ker P_C$ singularity but by the output-side conditions of Thm. 4.*

Remark (Provenance). Thm. 3 is elementary—the deterministic, two-sided cousin of the indirect problem of Dobrushin–Tsybakov, Wolf–Ziv, and Witsenhausen—and we claim structural, not technical, novelty. The constraint $\hat{U} \in V$ matters: enlarging reconstructions beyond the consumer’s reachable outputs could only lower the output-side function but is operationally meaningless, since every emitted $\hat{x}$ is read as $\mathcal{C}(\hat{x}) \in V$.

C. High resolution: the surrogate’s exact validity region

Apply Thm. 3 first, so the residual problem lives on the output with local metric $G(u)$ and D_Y is a locally quadratic non-difference distortion in the sense of Linder–Zamir [25]: $q_G(u, \hat{u}) := \frac{1}{2}(\hat{u} - u)^\top G(u)(\hat{u} - u) = \|W(u)(\hat{u} - u)\|^2$, $W = (G/2)^{1/2}$, $D_Y = q_G + O(\|\hat{u} - u\|^3)$.

Theorem 4 (High-resolution closed form; finite- D pinch). *Work in an intrinsic chart in which $U = \mathcal{C}(X)$ has an r -dimensional Lebesgue density with finite $h(U)$, taking h , G , and $\det$ in that chart, and assume the Linder–Zamir admissibility conditions [25] for (U, D_Y) : (a) $D_Y(u, \cdot) \in C^3$ with uniformly bounded third-order partials; (b) a unique minimum at $\hat{u} = u$ with a piecewise- C^1 , piecewise-invertible reproduction map; (c) $M(u) := \frac{1}{2} \nabla_u^2 D_Y(u, \hat{u})|_{\hat{u}=u} = G(u)/2 \succ 0$ a.e.; (d) the r -dimensional density just named; (e) $\mathbb{E}[\text{tr}(G(U)^{-1})] < \infty$; and take V relatively open, containing a neighborhood of $\text{supp}(U)$. Then:*

(i) As $D \rightarrow 0$,

$$R_C^{\text{true}}(D) = h(\mathcal{C}(X)) - \frac{r}{2} \log \frac{2\pi e D}{r} + \frac{1}{2} \mathbb{E} [\log \det(G(\mathcal{C}(X))/2)] + o(1),$$

the input-weighted Linder–Zamir asymptotic ($R(D) = h(X) - \frac{n}{2} \log \frac{2\pi e D}{n} + \mathbb{E} \log |\det W(X)|$), which we verified to machine precision against exact reverse water-filling in the fixed- G Gaussian case. For KL-type consumers G is the Fisher metric and the middle term is a Jeffreys-prior log-volume: rate = entropy of what the consumer sees + log-volume of the consumer’s metric + resolution. The chart is essential: the ambient output metric may be degenerate—softmax’s $G = \text{diag}(p) - pp^\top$ has rank $r = S - 1$, so $\det G = 0$ in $\mathbb{R}^S$ and the formula is meaningful only in the $(S - 1)$ -dimensional simplex chart, where it further needs $\text{supp}(U)$ bounded away from the boundary for (a),(g) to hold. The formula is chart-invariant as the combination $h(U) + \frac{1}{2} \mathbb{E} \log \det G$; genuinely curved U (a nonlinear submanifold) needs a manifold extension the 1999 theory does not supply (§B.4).

(ii) If $\alpha q_G \leq D_Y \leq \beta q_G$ on $\text{supp}(U) \times V$ —a global two-sided sandwich, which for a merely local surrogate generally holds only near the diagonal / on bounded support, and so pairs with the high-resolution regime where $(U, \hat{U})$ concentrate—then for every $D > 0$, $R_{q_G}(D/\alpha) \leq R_C^{\text{true}}(D) \leq R_{q_G}(D/\beta)$, by feasible-set inclusion; the bracket is computable at any D (Blahut–Arimoto; water-filling in the Gaussian case).

Remark (What coding to the surrogate certifies). Thms. 3–4 replace Cor. 2’s caveat (iii): coding to the point-dependent quadratic certifies the true downstream distortion *exactly* as $D \rightarrow 0$ and *within the* $[\alpha, \beta]$ pinch at any fixed D —the surrogate’s validity region is quantified, not asserted. Since d_C is an admissible general distortion with a reference letter, the finite-blocklength dispersion theory of Kostina–Verdú [26] should port under their regularity conditions.

D. What remains open, and a registered prediction

Genuinely open after Thms. 2–4: (a) the curved-image case (U on a submanifold without an ambient density—the high-resolution theory needs charts); (b) the consumer-relative *channel* coding theorem behind §VI; (c) the multi-observer rate region for $\{\mathcal{C}_k\}$, where the single-observer reduction fails because the observers’ quotients differ—the frontier flagged in §II; and (d) non-i.i.d. sources.

GO-6 (proposed; to be sealed before running). At matched rate R , three coders: (a) *output coding*—compress $\mathcal{C}(X)$ under D_Y (Thm. 3-optimal); (b) *surrogate coding*— X -space, d_{P_C} ; (c) *reconstruction coding*— X -space MSE. Predictions: $d_O(a) \leq d_O(b) \leq d_O(c)$, the (a)–(c) gap governed by the entropy share of $\ker P_C$ and strict whenever that share is positive, the (a)–(b) gap vanishing as R grows (Thm. 4). The synthetic harness of §IV suffices.

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a) *Reproducibility & provenance.*: All results resolve to committed registry entries and artifacts in the public companion repository [22]; each registry entry is committed *before* the run it governs, so the evidence that registration preceded experimentation is the *commit ordering* of the public history. We are explicit about its limits: git and hosting timestamps are not cryptographic third-party proof, and a snapshot taken *after* completion preserves the current history but does not retroactively establish timing. Stronger anchoring—an external immutable deposit (Zenodo/OSF) and any contemporaneous third-party timestamps (signed hashes, prior deposits)—accompanies the paper. Computational experimentation and drafting were carried out with an AI research assistant under the registration-first protocol of §X; every ledger row names its verifier and context-freshness.